

Question 1 (Easy):

A university's Training and Placement Cell have successfully completed its annual placement drive ~~com~~panies. The placement department has collected the salary packages (in LPA) offered to students by different companies. However, the records were entered in the order they were received, resulting in an unsorted list of package values.

Before publishing the placement statistics report, the department wants all package values arranged in ascending order. Since the university plans to handle placement data for thousands of students in the coming years, they want a sorting technique that is efficient for large datasets.

Your task is to implement the Merge sort algorithm to sort the package values in increasing order. Merge sort follows a divide-and-conquer approach by dividing the dataset into smaller subarrays, sorting them recursively, & then merging them back together.

The final sorted list will help the university identify minimum, median, & maximum packages while generating analytical reports.

Input

- First line contains an integer N , representing the number of placed students.

- Second line contains N space-separated integers, representing package values (multiplied by 100 to avoid decimals).

Output

print the package values in ascending order.

Constraints

- $1 \leq N \leq 100000$
- $100 \leq \text{package} \leq 10000$

Example

Input!

6

450 1200 800 650 300 1500

Output!

300 450 650 800 1200 1500

→ Logic!

- Merge sort uses the divide & conquer technique.

• `mergesort(arr, left, right)`

{

if (left < right)

{

mid = (left + right) / 2

mergesort(arr, left, mid)

mergesort(arr, mid + 1, right)

mergesort(arr, left, mid, right)

}

}


```
#include <iostream>
#include <vector>
using namespace std;

void merge(vector<int> &arr, int left, int mid, int right)
{
    int n1 = mid - left + 1;
    int n2 = right - mid;
    vector<int> L(n1), R(n2);

    for (int i = 0; i < n1; i++)
        L[i] = arr[left + i];

    for (int j = 0; j < n2; j++)
        R[j] = arr[mid + 1 + j];

    int i = 0, j = 0, k = left;
    while (i < n1 & j < n2)
    {
        if (L[i] <= R[j])
        {
            arr[k] = L[i];
            i++;
        }
        else
        {
            arr[k] = R[j];
            j++;
        }
        k++;
    }
    while (i < n1)
    {
        arr[k] = L[i];
        i++;
        k++;
    }
}
```

```
while (j < n2) {  
    arr[k] = r[j];  
    j++;  
    k++;  
}
```

```
}  
void mergesort(vector<int> &arr, int left, int right) {  
    if (left < right) {  
        int mid = left + (right - left) / 2;  
        mergesort(arr, left, mid);  
        mergesort(arr, mid + 1, right);  
        merge(arr, left, mid, right);  
    }  
}
```

```
int main() {
```

```
    int N;
```

```
    cin >> N;
```

```
    vector<int> arr(N);
```

```
    for (int i = 0; i < N; i++)
```

```
        cin >> arr[i];
```

```
    mergesort(arr, 0, N - 1);
```

```
    for (int i = 0; i < N; i++) {
```

```
        cout << arr[i];
```

```
        if (i != N - 1)
```

```
            cout << " ";
```

```
    }  
    return 0;
```

```
}
```

```

1 #include <iostream>
2 #include <vector>
3 using namespace std;
4
5 void merge(vector<int> &arr, int left, int mid, int right) {
6     int n1=mid-left+1;
7     int n2=right-mid;
8     vector<int> L(n1), R(n2);
9
10    for(int i=0;i<n1;i++)
11        L[i]=arr[left+i];
12
13    for(int j=0;j<n2;j++)
14        R[j]=arr[mid+1+j];
15
16    int i=0,j=0,k=left;
17    while (i<n1 && j<n2) {
18        if (L[i]<=R[j]) {
19            arr[k]=L[i];
20            i++;
21        } else {
22            arr[k]=R[j];
23            j++;
24        }
25        k++;
26    }
27    while (i<n1) {
28        arr[k]=L[i];
29        i++;
30        k++;
31    }
32    while (j<n2) {
33        arr[k]=R[j];
34        j++;
35        k++;
36    }
37 }
38 void mergeSort(vector<int> &arr, int left, int right) {
39     if (left<right) {
40         int mid=left+(right-left)/2;
41         mergeSort(arr, left, mid);
42         mergeSort(arr, mid + 1, right);
43         merge(arr, left, mid, right);
44     }
45 }
46 int main() {
47     int N;
48     cin >> N;
49     vector<int> arr(N);
50     for(int i=0;i<N;i++)
51         cin >> arr[i];
52     mergeSort(arr, 0, N - 1);
53     for (int i=0;i<N;i++) {
54         cout << arr[i];
55         if (i!= N - 1)
56             cout << " ";
57     }
58     return 0;
59 }

```

```

6
450 1200 800 650 300 1500
300 450 650 800 1200 1500

```


Question 2 (Medium):

A large e-commerce company tracks the time taken (in minutes) to process customer orders before shipment. Due to varying warehouse workload, processing times differ significantly from one order to another.

The analytics team wants to study operational efficiency by sorting all processing times & generating statistical information. Since millions of orders are processed every month, the company requires an efficient sorting technique with predictable performance.

Your task is to use Merge Sort to arrange all processing times in ascending order.

After sorting algorithm, perform the following operations:

1. Display the sorted processing times.
2. Find the median processing time.
3. Count the number of orders whose processing time is greater than the median.
4. Display the difference between the fastest & slowest processing time.

The solution should efficiently handle large datasets & demonstrate a clear understanding of Merge Sort's divide-&-conquer strategy.

Input:

- First line contains integer N .
- Second line contains N space-separated processing times.

Output:

- Sorted processing times.
- Median processing time.
- Count of orders greater than median.
- Difference between maximum & minimum processing time.

Constraints

- $1 \leq N \leq 100000$
- $1 \leq \text{Time} \leq 100000$

Example

Input:

7

12 25 18 30 15 22 40

Output:

12 15 18 22 25 30 40

Median: 22

Orders Above Median: 3

Difference: 28

→ Logic:

- Read N & processing times.
- Sort array using Merge sort.
- Print sorted array.
- Find median:
 - if N is odd $\rightarrow$ middle element
 - if N is even $\rightarrow$ average of two middle elements (integer division)
- Count how many processing times are greater than median.
- Compute the difference.


```

#include <iostream>
#include <vector>
using namespace std;

void merge(vector<int> &all, int left, int mid,
int right) {
    int n1 = mid - left + 1;
    int n2 = right - mid;
    vector<int> L(n1), R(n2);

    for (int i = 0; i < n1; i++)
        L[i] = all[left + i];

    for (int j = 0; j < n2; j++)
        R[j] = all[mid + 1 + j];

    int i = 0, j = 0, k = left;

    while (i < n1 & j < n2) {
        if (L[i] <= R[j])
            all[k++] = L[i++];
        else
            all[k++] = R[j++];
    }

    while (i < n1)
        all[k++] = L[i++];

    while (j < n2)
        all[k++] = R[j++];
}

```



```
void mergesort (vector<int> &all, int left, int right) {
    if (left < right) {
        int mid = left + (right - left) / 2;
        mergesort (all, left, mid);
        mergesort (all, mid + 1, right);
        merge (all, left, mid, right);
    }
}
```

```
int main () {
    int N;
    cin >> N;
    vector<int> all (N);
    for (int i = 0; i < N; i++)
        cin >> all[i];
    mergesort (all, 0, N - 1);

    for (int i = 0; i < N; i++)
        cout << all[i] << " ";
    cout << endl;
    int median;
    if (N % 2 == 1)
        median = all [N / 2];
    else
        median = (all [N / 2 - 1] + all [N / 2]) / 2;
    cout << "Median: " << median << endl;
    int count = 0;
    for (int i = 0; i < N; i++) {
        if (all [i] > median)
            count++;
    }
```

```
    cout << "Orders Above Median: " << count << endl;
    int difference = all [N - 1] - all [0];
    cout << "Difference: " << difference << endl;
}
```

```

5 void merge(vector<int> &arr, int left, int mid, int right) {
6     int n1=mid-left+1;
7     int n2=right-mid;
8
9     vector<int> L(n1), R(n2);
10
11     for(int i=0;i<n1;i++)
12         L[i]=arr[left + i];
13
14     for(int j=0;j<n2;j++)
15         R[j]=arr[mid + 1 + j];
16
17     int i=0,j=0,k=left;
18
19     while(i<n1 && j<n2) {
20         if (L[i] <= R[j])
21             arr[k++]=L[i++];
22         else
23             arr[k++]=R[j++];
24     }
25
26     while(i < n1)
27         arr[k++] = L[i++];
28
29     while(j < n2)
30         arr[k++] = R[j++];
31 }
32
33 void mergeSort(vector<int> &arr, int left, int right) {
34     if (left < right) {
35         int mid = left + (right - left) / 2;
36         mergeSort(arr, left, mid);
37         mergeSort(arr, mid + 1, right);
38
39         merge(arr, left, mid, right);
40     }
41 }
42
43 int main() {
44     int N;
45     cin >> N;
46     vector<int> arr(N);
47
48     for(int i=0;i<N;i++)
49         cin >> arr[i];
50
51     mergeSort(arr, 0, N - 1);
52
53     for(int i=0;i<N;i++)
54         cout << arr[i] << " ";
55     cout << endl;
56
57     int median;
58     if(N % 2 == 1)
59         median = arr[N / 2];
60     else
61         median=(arr[N / 2 - 1] + arr[N / 2]) / 2;
62
63     cout << "Median: " << median << endl;
64
65     int count = 0;
66     for(int i=0;i<N;i++) {
67         if(arr[i] > median)
68             count++;
69     }
70
71     cout << "Orders Above Median: " << count << endl;
72     int difference = arr[N - 1] - arr[0];
73
74     cout << "Difference: " << difference << endl;
75     return 0;

```

```

7
12 25 18 30 15 22 40
12 15 18 22 25 30 40
Median: 22
Orders Above Median: 3
Difference: 28

```



Topic 2: Quick Sort

Question 1 (Easy):

A cloud infrastructure company manages hundreds of servers distributed across different geographical locations. Every hour, each server reports its average response time in milliseconds. The collected data is stored in the order it arrives & is not sorted.

To generate performance reports, the operations team wants the response times arranged from the fastest server to the slowest server. Since the data can be very large, an efficient in-place sorting technique is required.

Your task is to implement Quick Sort the response times in ascending order. Built-in sorting methods are not allowed.

After sorting, the operations team will use the report to identify high-latency servers & optimize system performance.

Input:

- First line contains integer N .
- Second line contains N space-separated response times.

Output:

Print the response times in ascending order.

Constraints:

- $1 \leq N \leq 100000$
- $1 \leq \text{Response Time} \leq 10000$

Example:Input:

6

120 45 78 200 60 95

Output:

45 60 78 95 120 200

→ Logic:

QuickSort (all, low, high)

↓

if (low < high)

↓

pivot = partition (all, low, high)

QuickSort (all, low, pivot - 1)

QuickSort (all, pivot + 1, high)

}

}


```
#include <iostream>
#include <vector>
using namespace std;
```

```
int partition (vector<int> &all, int low, int high)
{
    int pivot = all[high];
    int i = low - 1;
    for (int j = low; j < high; j++)
    {
        if (all[j] < pivot)
        {
            i++;
            swap(all[i], all[j]);
        }
    }
    swap(all[i+1], all[high]);
    return i+1;
}
```

```
void quicksort (vector<int> &all, int low, int high)
{
    if (low < high)
    {
        int pi = partition(all, low, high);
        quicksort(all, low, pi-1);
        quicksort(all, pi+1, high);
    }
}
```

```
int main()
{
    int N;
    cin >> N;
    vector<int> all(N);
    for (int i = 0; i < N; i++)
    {
        cin >> all[i];
    }
    quicksort(all, 0, N-1);
    for (int i = 0; i < N; i++)
    {
        cout << all[i] << " ";
    }
    return 0;
}
```



```
1 #include <iostream>
2 #include <vector>
3 using namespace std;
4
5 int partition(vector<int> &arr, int low, int high) {
6     int pivot=arr[high];
7     int i=low-1;
8
9     for(int j=low;j<high;j++) {
10         if(arr[j] < pivot) {
11             i++;
12             swap(arr[i], arr[j]);
13         }
14     }
15     swap(arr[i + 1], arr[high]);
16     return i + 1;
17 }
18
19 void quickSort(vector<int> &arr, int low, int high) {
20     if(low < high) {
21         int pi=partition(arr, low, high);
22         quickSort(arr, low, pi - 1);
23         quickSort(arr, pi + 1, high);
24     }
25 }
26
27 int main() {
28     int N;
29     cin >> N;
30     vector<int> arr(N);
31
32     for(int i=0;i<N;i++)
33         cin >> arr[i];
34     quickSort(arr, 0, N - 1);
35
36     for(int i=0;i<N;i++)
37         cout << arr[i] << " ";
38
39     return 0;
40 }
```

```
6
120 45 78 200 60 95
45 60 78 95 120 200
```

Question 2 (medium):

A financial analytics company receives daily trade-value data from multiple stock exchanges. Analysts use this information to identify high-value trading activity & market trends.

The trade values are received in random order throughout the day. Before analysis can begin, the data must be sorted in descending order.

Your task is to implement Quick Sort to arrange the trade values from highest to lowest.

After sorting, perform the following operations:

1. Display the sorted trade values.
2. Find & the top 5 highest trade values.
3. Calculate the average of the top 5 values.
4. Count how many trade values are greater than the overall average of all trade values.

This problem evaluates your understanding of partition recursion, & real-world data analysis using Quick Sort.

Input:

- First line contains integer n .
- Second line contains n space-separated trade values.

Output:

- Sorted trade values (descending).
- Top 5 values.
- Average of Top 5 values.

- Count of values greater than overall average.

Constraints:

- $5 \leq N \leq 100000$
- $1 \leq \text{Trade value} \leq 10^5$

Example:

Input:

8

500 1200 700 2000 900 1500 3000 2500

Output:

3000 2500 2000 1500 1200 900 700 500

Top 5: 3000 2500 2000 1500 1200

Average of Top 5: 2040

Values Above Overall Average: 4.

→ Logic:

- Read N & the trade values.
- Use quick sort to sort array in descending order.
- Print the sorted values.
- Take the first 5 elements as the Top 5 values.
- Compute the average of the Top 5 values.
Average = (Top5 sum) / 5
- Compute overall average
Overall Average = Total sum / N
- Count how many trade values are greater than the overall average.
- Print all required results.


```
#include <iostream>
```

```
#include <vector>
```

```
using namespace std;
```

```
int partition (vector<long long> &arr, int low, int high) {
    long long pivot = arr[high];
    int i = low - 1;
```

```
    for (int j = low; j < high; j++) {
        if (arr[j] > pivot) {
            i++;
            swap(arr[i], arr[j]);
        }
    }
```

```
    swap(arr[i+1], arr[high]);
    return i+1;
```

```

}
void quickSort (vector<long long> &arr, int low, int high) {
    if (low < high) {
        int pi = partition (arr, low, high);
        quickSort (arr, low, pi-1);
        quickSort (arr, pi+1, high);
    }
}
```

```
int main () {
```

```
    int n;
```

```
    cin >> n;
```

```
    vector<long long> arr(n);
```

```
    long long totalSum = 0;
```

```
    for (int i = 0; i < n; i++) {
```

```
        cin >> arr[i];
```

```
totalsum += arr[i];
```

```
} quicksort(arr, 0, n-1);  
for (int i=0; i<n; i++)  
    cout << arr[i] << " ";  
cout << endl;
```

```
cout << "Top 5:";  
long long topsum = 0;  
for (int i=0; i<5; i++) {  
    cout << arr[i] << " ";  
    topsum += arr[i];  
}
```

```
cout << endl;  
cout << "Average of Top 5:" << topsum/5 << endl;
```

```
double overallAvg = (double) totalsum / n;
```

```
int count = 0;  
for (int i=0; i<n; i++) {  
    if (arr[i] > overallAvg)  
        count++;  
}
```

```
cout << "Values Above Overall Average:" << count << endl;  
return 0;
```

```
}
```



```

1 #include <iostream>
2 #include <vector>
3 using namespace std;
4
5 int partition(vector<long long> &arr, int low, int high) {
6     long long pivot=arr[high];
7     int i=low-1;
8
9     for(int j=low;j<high;j++) {
10         if(arr[j] > pivot) {
11             i++;
12             swap(arr[i], arr[j]);
13         }
14     }
15     swap(arr[i + 1], arr[high]);
16     return i + 1;
17 }
18
19 void quickSort(vector<long long> &arr, int low, int high) {
20     if(low < high) {
21         int pi=partition(arr, low, high);
22         quickSort(arr, low, pi - 1);
23         quickSort(arr, pi + 1, high);
24     }
25 }
26
27 int main() {
28     int N;
29     cin >> N;
30     vector<long long> arr(N);
31
32     long long totalSum=0;
33     for(int i=0;i<N;i++) {
34         cin >> arr[i];
35         totalSum += arr[i];
36     }
37     quickSort(arr, 0, N - 1);
38     for(int i=0;i<N;i++) {
39         cout << arr[i] << " ";
40     }
41     cout << endl;
42
43     long long topSum=0;
44     for(int i=0;i<5;i++) {
45         cout << arr[i] << " ";
46         topSum += arr[i];
47     }
48     cout << endl;
49     cout << "Average of Top 5: " << topSum / 5 << endl;
50
51     double overallAvg = (double)totalSum / N;
52
53     int count = 0;
54     for(int i=0;i<N;i++) {
55         if (arr[i] > overallAvg)
56             count++;
57     }
58     cout << "Values Above Overall Average: " << count << endl;
59     return 0;
60 }

```

```

8
500 1200 700 2000 900 1500 3000 2500
3000 2500 2000 1500 1200 900 700 500
Top 5: 3000 2500 2000 1500 1200
Average of Top 5: 2040
Values Above Overall Average: 3

```

Topic 3: Heap Sort

Question 1 (Easy):

An online gaming platform is hosting a national-level tournament involving thousands of participants. At the end of the tournament, player scores are collected from multiple game servers & stored in random order.

Before announcing the final rankings, the organizers need all scores sorted in ascending order. Since the leaderboard data can be very large, they have decided to use Heap Sort, which provides guaranteed $O(N \log N)$ performance.

Your task is to implement Heap Sort & arrange all player scores in increasing order.

The sorted output will help tournament officials generate rankings & determine qualification thresholds for future events.

Input:

- First line contains integer N .
- Second line contains N space-separated player scores.

Output:

print the scores in ascending order.

Constraints:

- $1 \leq N \leq 100000$
- $0 \leq \text{score} \leq 1000000$

Example:

Input:

6

850 1200 950 600 1500 1000

Output:

600 850 950 1000 1200 1500

→ Logic:

- Read N & the array.
- Build a Max Heap by calling `heapify()` from the last non-leaf node to the root.
- Repeat until the heap size becomes 1:
Swap first element with the last element.
Decrease the heap size.
Call `heapify()` on the root.
- Print the sorted array.

Pseudocode

Heapsort (arr)

Build Max Heap

for $i = n-1$ to 1

 swap(arr[0], arr[i])

 heapify(arr, i, 0)

Print array.

```
#include <iostream>
#include <vector>
using namespace std;

void heapify (vector<int> &arr, int n, int i) {
    int largest = i;
    int left = 2 * i + 1;
    int right = 2 * i + 2;

    if (left < n && arr[left] > arr[largest])
        largest = left;

    if (right < n && arr[right] > arr[largest])
        largest = right;

    if (largest != i) {
        swap(arr[i], arr[largest]);
        heapify(arr, n, largest);
    }
}
```

```
void heapSort (vector<int> &arr) {
    int n = arr.size();
    for (int i = n/2 - 1; i >= 0; i--)
        heapify(arr, n, i);

    for (int i = n - 1; i > 0; i--) {
        swap(arr[0], arr[i]);
        heapify(arr, i, 0);
    }
}
```



```
int main() {
    int N;
    cin >> N;
```

```
    vector<int> scores(N);
```

```
    for (int i=0; i<N; i++)
        cin >> scores[i];
    heapSort(scores);
```

```
    for (int i=0; i<N; i++)
        cout << scores[i] << " ";
```

```
    return 0;
```

```
}
```




```
1 #include <iostream>
2 #include <vector>
3 using namespace std;
4
5 void heapify(vector<int>& arr, int n, int i) {
6     int largest=i;
7     int left=2*i+1;
8     int right=2*i+2;
9
10    if(left < n && arr[left] > arr[largest])
11        largest=left;
12
13    if(right < n && arr[right] > arr[largest])
14        largest=right;
15
16    if(largest != i) {
17        swap(arr[i], arr[largest]);
18        heapify(arr, n, largest);
19    }
20 }
21
22 void heapSort(vector<int>& arr) {
23     int n=arr.size();
24     for(int i=n/2-1;i>=0;i--)
25         heapify(arr, n, i);
26
27     for(int i=n-1;i>0;i--) {
28         swap(arr[0], arr[i]);
29         heapify(arr, i, 0);
30     }
31 }
32
33 int main() {
34     int N;
35     cin >> N;
36
37     vector<int> scores(N);
38
39     for(int i=0;i<N;i++)
40         cin >> scores[i];
41     heapSort(scores);
42
43     for(int i=0;i<N;i++)
44         cout << scores[i] << " ";
45     return 0;
```

6

850 1200 950 600 1500 1000

600 850 950 1000 1200 1500



Page No. 45
Date / /

Question 2 (Medium):

A multi-specialty hospital maintains records of emergency response times for all critical cases handled during a month. Each response time represents the number of minutes taken by the emergency team to reach and begin treatment for a patient.

Hospital administrators want to analyze the efficiency of the services. To do so, all response times must first be sorted using Heap Sort.

After sorting the response times in ascending order, generate the following statistics:

1. Display the sorted response times.
2. Find the fastest response time.
3. Find the slowest response time.
4. Calculate the average response time.
5. Count the number of cases handled faster than the average.
6. Calculate the percentage of cases handled faster than the average.

The hospital intends to use this information to improve resource allocation & reduce emergency response delays.

Input:

- First line contains integer N .
- Second line contains N space-separated response times.

Output:

- Sorted response times.
- Fastest response time.
- Slowest response time.
- Average response time.
- Number of cases faster than average.
- Percentage of cases faster than average.

Constraints:

- $1 \leq N \leq 1000000$
- $1 \leq \text{Time} \leq 500$

Example

Input:

8

12 7 15 9 20 5 10 8

Output:

5 7 8 9 10 12 15 20

Fastest: 5

Slowest: 20

Average: 10.75

Cases Faster than Average: 5

Percentage: 62.50%

→ Logic:

- Read N emergency response times.
- Build a max heap using `heapify()`.
- Perform heap sort to arrange the response in ascending order.
- $\text{Percentage} = \text{Count} / N \times 100$
- Print all required statistics.


```
#include <iostream>
#include <vector>
#include <algorithm>
using namespace std;
```

```
void heapify (vector<int> &arr, int n, int i) {
    int largest = i;
    int left = 2 * i + 1;
    int right = 2 * i + 2;
```

```
    if (left < n & arr[left] > arr[largest])
        largest = left;
```

```
    if (right < n & arr[right] > arr[largest])
        largest = right;
```

```
    if (largest != i) {
        swap(arr[i], arr[largest]);
        heapify(arr, n, largest);
    }
```

```
void heapsort (vector<int> &arr)
```

```
{
    int n = arr.size();
    for (int i = n / 2 - 1; i >= 0; i--)
        heapify(arr, n, i);
```

```
    for (int i = n - 1; i > 0; i--)
```

```
    {
        swap(arr[0], arr[i]);
        heapify(arr, i, 0);
    }
```

```
}
```

```

int main() {
    int N;
    cin >> N;
    vector<int> response(N);

```

```

    double sum = 0;
    for (int i = 0; i < N; i++) {
        cin >> response[i];
        sum += response[i];
    }

```

3

```

    heap_sort(response);

```

```

    for (int i = 0; i < N; i++)
        cout << response[i] << " ";
    cout << endl;

```

```

    cout << "fastest: " << response[0] << endl;
    cout << "slowest: " << response[N-1] << endl;

```

```

    double average = sum / N;
    cout << fixed << setprecision(2);
    cout << "Average: " << average << endl;

```

```

    int count = 0;
    for (int i = 0; i < N; i++) {
        if (response[i] < average)
            count++;
    }

```

3

```

    cout << "cases faster than Average: " << count << endl;
    double percentage = (count * 100.0) / N;
    cout << "percentage: " << percentage << "%." << endl;
    return 0;

```

3

```

1 #include <iostream>
2 #include <vector>
3 #include <iomanip>
4 using namespace std;
5
6 void heapify(vector<int> &arr, int n, int i)
7 {
8     int largest=i;
9     int left=2*i+1;
10    int right=2*i+2;
11
12    if(left < n && arr[left] > arr[largest])
13        largest=left;
14
15    if(right < n && arr[right] > arr[largest])
16        largest=right;
17
18    if(largest != i)
19    {
20        swap(arr[i], arr[largest]);
21        heapify(arr, n, largest);
22    }
23 }
24 void heapSort(vector<int> &arr)
25 {
26     int n=arr.size();
27     for(int i=n/2-1;i>=0;i--)
28         heapify(arr, n, i);
29
30     for(int i=n-1;i>=0;i--)
31     {
32         swap(arr[0], arr[i]);
33         heapify(arr, i, 0);
34     }
35 }
36 int main()
37 {
38     int N;
39     cin >> N;
40     vector<int> response(N);
41     double sum=0;
42
43     for(int i=0;i<N;i++)
44     {
45         cin >> response[i];
46         sum += response[i];
47     }
48     heapSort(response);
49
50     for(int i=0;i<N;i++)
51     {
52         cout << response[i] << " ";
53     }
54     cout << endl;
55
56     cout << "Fastest: " << response[0] << endl;
57     cout << "Slowest: " << response[N - 1] << endl;
58
59     double average=sum / N;
60     cout << fixed << setprecision(2);
61     cout << "Average: " << average << endl;
62
63     int count=0;
64     for(int i=0;i<N;i++)
65     {
66         if (response[i] < average)
67             count++;
68     }
69     cout << "Cases Faster Than Average: " << count << endl;
70
71     double percentage = (count * 100.0) / N;
72     cout << "Percentage: " << percentage << "%" << endl;
73     return 0;
74 }

```

```

8
12 7 15 9 20 5 10 8
5 7 8 9 10 12 15 20
Fastest: 5
Slowest: 20
Average: 10.75
Cases Faster Than Average: 5
Percentage: 62.50%

```